BEFA UNIT III

PRODUCTION

Production is the transformation or conversion of resources into commodities over time. Economists view production as an activity through which utility is created or enhanced for a product. A firm is a business unit which undertakes the activity of transforming inputs into output of goods and services.



FACTORS OF PRODUCTION

Factors of production is an economic term that describes the inputs that are used in the production of goods or services in order to make an economic profit. The factors of production include land, labor, capital and entrepreneurship. These production factors are also known as management, machines, materials and labor, and knowledge has recently been talked about as a potential new factor of production.

1. Land

Land is short for all the natural resources available to create supply. It includes raw land and anything that comes from the land. It can be a non-renewable resource.

That includes commodities such as oil and gold. It can also be a renewable resource, such as timber. Once man changes it from its original condition, it becomes a capital good. For example, oil is a natural resource, but gasoline is a capital good. Farmland is a natural resource, but a shopping center is a capital good.

The income earned by owners of land and other resources is called rent.

2. Labour

Labor is the work done by people. The value of labor depends on workers' education, skills, and motivation. It also depends on productivity. That measures how much each hour of worker time produces in output.

The reward or income for labor is wages.

3. Capital

Capital is short for capital goods. These are man-made objects like machinery, equipment, and chemicals, that are used in production. That's what differentiates them from consumer goods. For example, capital goods include industrial and commercial buildings, but not private housing. A commercial aircraft is a capital good but a private jet is not.

The income earned by owners of capital goods is called interest.

4. Entrepreneurship

Entrepreneurship is the drive to develop an idea into a business. An entrepreneur combines the other three factors of production to add to supply. The most successful are innovative risk-takers.

The income entrepreneurs earn is profits.

PRODUCTION FUNCTION

The production function expresses a functional relationship between physical inputs and physical outputs of a firm at any particular time period. The output is thus a function of inputs. So, production function is an input – output relationship. Mathematically production function can be written as

$$Q = f(L_1, L_2\ C, O, T)$$

Here output is the function of inputs. Hence output becomes the dependent variable and inputs are the independent variables.

Q = Output
f = Function of
L_1 = Land
L_2 = Labour
C = Capital
O = Organization
T = Technology

Definition :

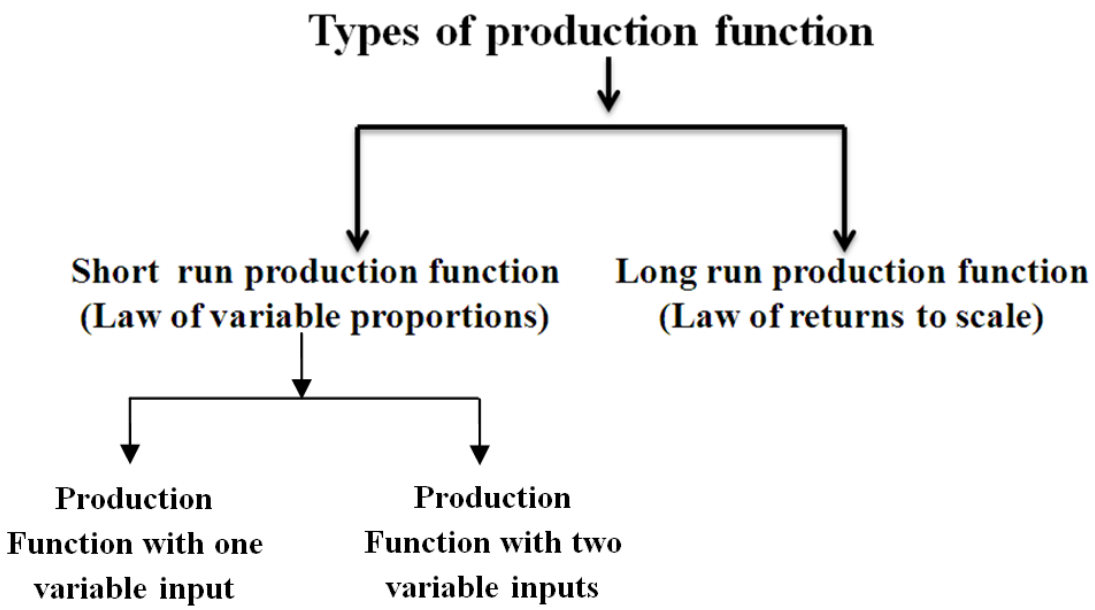
Samueson defines the production function as "The technical relationship which reveals the maximum amount of output capable of being produced by each and every set of inputs"

Michael R Baye defines the production function as" That function which defines the maximum amount of output that can be produced with a given set of inputs."

Assumptions:

Production function has the following assumptions.

1. The production function is related to a particular period of time.
2. There is no change in technology.
3. The producer is using the best techniques available.
4. The factors of production are divisible.
5. Production function can be fitted to a short run or to long run.



BEFA UNIT III

PRODUCTION FUNCTION WITH ONE VARIABLE INPUT

The law of variable proportions which was earlier called as "Law of diminishing returns has played a vital role in the modern economics theory. Assume that a firms' production function consists of fixed quantities of all inputs (land, equipment, etc.) except labour which is a variable input. If you go on adding the variable input, say, labor, the total output in the initial stages will increase at an increasing rate, and after reaching certain level of output the total output will increase at declining rate. If variable factor inputs are added further to the fixed factor input, the total output may decline. This law is of universal nature and it proved to be true in agriculture.

Assumptions of the Law: The law is based upon the following assumptions:

1. Only one factor is varied
2. The scale of output is unchanged
3. The technique of production is unchanged
4. All units of factor input varied are homogeneous

Three stages of law:

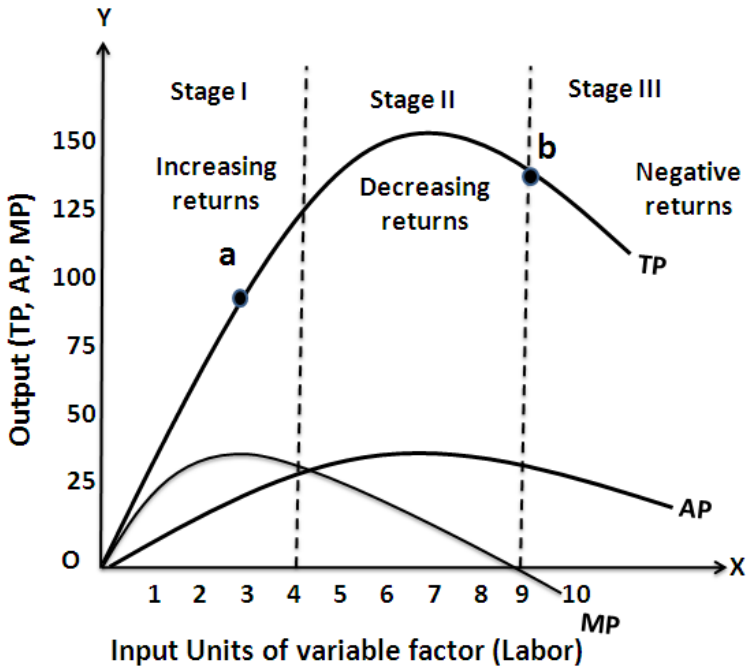
The behaviour of the Output when the varying quantity of one factor is combined with a fixed quantity of the other can be divided in to three district stages. The three stages can be better understood by following the table.

Production function with one variable input

Fixed Factor (FF) (land)	Variable Factor (VF) (labor)	Total Product (TP)	Average Product (AP)	Marginal Product (MP)	
1	1	20	20	20	
1	2	50	25	30	STAGE I
1	3	90	30	40	
1	4	120	30	30	
1	5	135	27	15	
1	6	144	24	9	STAGE II
1	7	147	21	3	
1	8	148	18.5	1	
1	9	148	16.4	0	
1	10	145	14.5	- 3	STAGE III

To clarify the relationship, the following are measurements of product.

1. **Total product(TP)**:- Means the total number of units of output produced per unit of time by all factor inputs.
2. **Average product(AP)**:- Is obtained by dividing the total product by the total units of variable factor.
3. **Marginal product(MP):-** is defined as the change in the total product per unit change in the variable input.



	I	II	III
TP	Increases at an increasing rate	Increases at a decreasing rate and becomes maximum	Decreases
AP	Increases and reaches maximum	Decreases	Continuous to decrease
MP	Increases and reaches maximum	Continuous to fall	Becomes negative

From the above graph the law of variable proportions operates in three stages. In the first stage, total product increases at an increasing rate. The marginal product in this stage increases at an increasing rate resulting in a greater increase in total product. The average product also increases. This stage continues up to the point where average product is equal to marginal product. The law of increasing returns is in operation at this stage. The law of diminishing returns starts operating from the second stage onwards. At the second stage total product increases only at a diminishing rate. The average product also declines. The second stage comes to an end where total product becomes maximum and marginal product becomes zero. The marginal product becomes negative in the third stage. So the total product also declines. The average product continues to decline.

We can sum up the above relationship thus when 'AP' is rising, "MP' rises more than "AP; When 'AP" is maximum and constant, 'MP' becomes equal to 'AP' when 'AP' starts falling, 'MP' falls faster than 'AP'.

Thus, the total product, marginal product and average product pass through three phases, viz., increasing diminishing and negative returns stage. The law of variable proportion is nothing but the combination of the law of increasing and demising returns.

PRODUCTION FUNCTION WITH TWO VARIABLE INPUTS

Isoquants analyse and compare the different combinations of capital & labour and output. The term isoquant has its origin from two words "iso" and "quantus". 'iso' is a Greek word meaning 'equal' and 'quantus' is a Latin word meaning 'quantity'. Isoquant therefore, means equal quantity. An isoquant curve is therefore called as iso-product curve or equal product curve or production indifference curve.

Thus, an isoquant shows all possible combinations of two inputs, which are capable of producing equal or a given level of output. Since each combination yields same output, the producer becomes indifferent towards these combinations.

Assumptions:

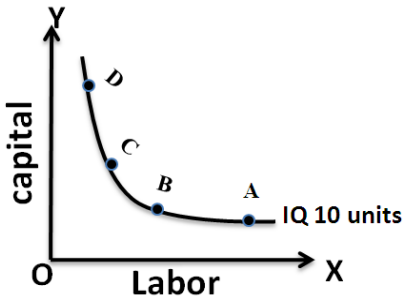
1. There are only two factors of production, viz. labour and capital.
2. The two factors can substitute each other up to certain limit
3. The shape of the isoquant depends upon the extent of substitutability of the two inputs.
4. The technology is given over a period.

For example:- Now the firm can combine labor and capital in different proportions and can maintain specified level of output say, 10 units of output of a product X. It may combine alternatively as follows:

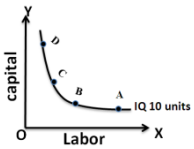
In the below table, combination 'A' represent 1 unit of capital and 10 units of labour and produces '10' units of a product. All other combinations in the table are assumed to yield the same given output of a product say '10' units by employing any one of the alternative combinations of the two factors labour and capital. If we plot all these combinations on a paper and join them, we will get a curve called Iso-quant curve as shown below.

Labour is on the X-axis and capital is on the Y-axis. IQ is the Iso-Quant curve which shows all the alternative combinations A, B, C, D which can produce 10 units of a product.

Combination	Capital	Labor	output
A	1	10	10 units
B	2	6	10 "
C	3	3	10 "
D	4	1	10 "



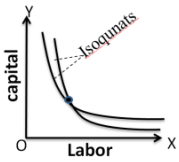
Features of an ISOQUANT:



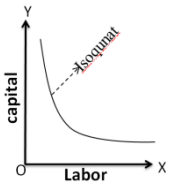
1. **Downward sloping:-**If one of the inputs is reduced, the other input has to be increased. There is no question of increase in both the inputs to yield a given output.



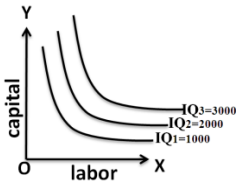
2. **Don't touch the axes:-** The isoquant touches neither X-axis nor Y-axis, as both inputs are required to produce a given product.If an isoquant is touching the X-axis, it means output is possible even by using a factor(Ex: Labor alone without using capital). But, this is unrealistic.



3. **Don't intersect:-** Iso-quants representing different levels of output never intersect or touch or be tangent to each other. If they intersect to each other, they have a common point on them which means that the same amount of labor and capital produce two different levels of output.



4. **Convex to origin:-**Isoquants are convex to the origin. It is because the inputs factor are not perfect substitutes. One input factor is substituted by other input factor in a decreasing marginal rate.The convexity of isoquant suggests that MRTS is diminishing which means that as quantities of one factor-labor is increased , the less of another factor-capital will be given up, if output level is to be kept constant.



5. **Upper isoquants represent higher level of output**:- Each isoquant represents a different quantity of output. Higher isoquants indicate a higher level of output.

LAW OF RETURNS TO SCALE

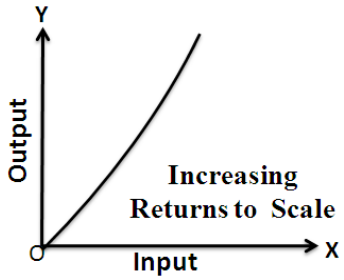
The concept of variable proportions is a short-run phenomenon as in these period fixed factors cannot be changed and all factors cannot be changed. On the other hand in the long-term all factors can be changed as made variable. When we study the changes in output when all factors or inputs are changed, we

study returns to scale. An increase in the scale means that all inputs or factors are increased in the same proportion. In variable proportions, the cooperating factors may be increased or decreased and one faster (Ex. Land in agriculture (or) machinery in industry) remains constant so that the changes in proportion among the factors result in certain changes in output. In returns to scale, all the necessary factors or production are increased or decreased to the same extent so that whatever the scale of production, the proportion among the factors remains the same.

Assumptions

1. Technique of production is unchanged
2. All units of factors are homogeneous
3. Returns are measured in physical terms

When a firm expands, its scale increases all its inputs proportionally, then technically there are three possibilities.



1. Law of increasing returns to scale:- if a proportionate/percentage increase in the output is larger than the proportionate/percentage increase in inputs, there are increasing returns.

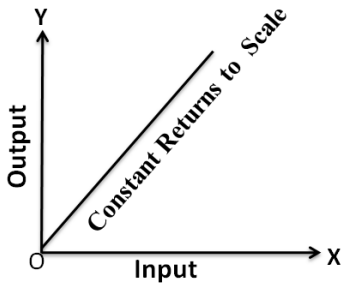
For example: If a 5% increase in inputs, results in 10% increase in the output, a firm is said to attain increased returns.

$$\text{Production Factor Co-efficient} = \frac{\%\ \text{Change in output}}{\%\ \text{Change in input}}$$

If PFC > 1, it means increasing returns to scale

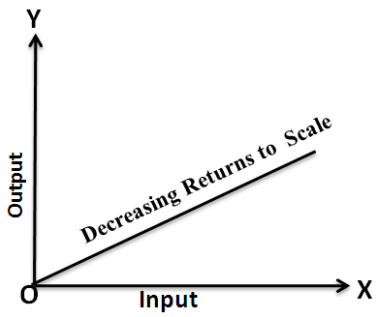
If PFC = 1, it means constant returns to scale

If PFC < 1, it means decreasing returns to scale



2. Law of constant returns to scale:- if the proportionate increase in all the inputs is equal to the proportionate increase in output, then situation of constant returns to scale occurs.

For Example:- If the inputs are increased at 10% and if the resultant output also increases a 10% then the organization is said to achieve constant returns to scale.



3. Law of decreasing returns to scale:- if the proportionate increase in output is less than the proportionate increase in input, then a situation of decreasing returns to scale occurs.

For Example:- If the inputs are increased by 10% and if the resultant output increases only by 5% then the organization is said to achieve decreasing returns to

scale.

the following table illustrates these laws clearly.

Capital (in units)	Labor (in units)	% of increase in both inputs	Output (in units)	% of increase in output	Laws applicable
1	3	-	50	-	-
2	6	100	120	140	Law of increasing returns to scale
4	12	100	240	100	Law of constant returns to scale
8	24	100	360	50	Law of decreasing returns to scale

From the above table it is clear that with 1 unit of capital and 3 units of labor, the firm produces 50 units of output. When the inputs are doubled 2 units of capital and 6 units of labor, the output has gone up to 120 units. Thus when inputs are increased by 100%, the output has increased by 140%. That is , output has increased by more than double. This is governed by **law of increasing returns to scale**.

When the inputs are further doubled that is to 4 units of capital and 12 units of labor, the output has gone to 240 units. Thus, when inputs are increased by 100%., the output has increased by 100%. That is, output has doubled. This governed by **law of constant returns to scale.**

When the inputs further doubled, that is, to 8 units of capital and 24 units of labor, the output has gone up to 360 units. Thus, when inputs are increased by 100%, the output has increased by only 50%. This is governed by **law of decreasing returns to scale.**

DIFFERENT TYPES OF PRODUCTION FUNCTIONS

1. Cobb-Douglas Production Function

This production function was proposed by C. W. Cobb and P. H. Dougles. This famous statistical production function is known as Cobb-Douglas production function. Originally the function is applied on the empirical study of the American manufacturing industry from 1899 to 1922. Cobb – Douglas production function takes the following mathematical form.

$$Q = aL^bK^C$$

$$Q = 1.01L^{0.75}K^{0.25}$$

The above production function shows that 1% change in labor input, capital remaining the same, is associated with a 0.75 percent change in output. Similarly, 1% change in capital, labor remaining the same, is associated with a 0.25 percent change in output.

Q = Total production/Output
a = Total factor productivity/Multi -factor productivity
(The ratio of output to the sum of associated labor and capital inputs). Productivity is measure of efficiency of labor/machines etc, in converting input into output.

$$\text{Factor productivity} = \frac{\text{Ouput}}{\text{Factor input}}$$

$$\text{Total Factor Productivity} = \frac{\text{Total Ouput}}{\text{Total inputs}}$$

L = Index of employment of labor = Labour units
K = Index of employment of capital = Capital units
b = Output elasticity of labor (1% change in labor results in b% change in output)
c = Output elasticity of capital (1% change in capital results in c% change in output)

Assumptions:

It has the following assumptions

1. The function assumes that output is the function of two factors viz. capital and labour.
2. It is a linear homogenous production function of the first degree
3. There are constant returns to scale (b+c=1)
4. All inputs are homogenous
5. There is perfect competition
6. There is no change in technology
7. Both L&K should be positive for to Q to exist. If either of these is zero, Q will be zero. This implies that both labor and capital will be combined to get output.

Simple Cobb-Douglas example:

$$Q = aL^bK^c$$
$$Q = 2L^{0.5}K^{0.5}$$

Let K=100, L=25

$$Q = 2x100^{0.5}x25^{0.5}$$
$$= 2x10x5$$
$$= 100 \text{ units}$$

1% increase in labor results in 0.5% increase in output:
Labor units after 1% increase = 100+(100 x1/100) = 101 units

$$Q = 2L^{0.5}K^{0.5}$$
$$Q = 2 \times 101^{0.5} \times 25^{0.5}$$
$$Q = 2 \times 10.0498 \times 5$$
$$Q = 100.498 = 100.50$$

$$\% \text{ increase in output} = \frac{100.50-100}{100} x100 = 0.50\%$$

2. Leontief Production Function:

Leontief production function, also known as **Fixed Proportion Production Function,** uses fixed proportion of inputs having no substitutability between them. It is regarded as the limiting case for constant elasticity of substitution.

The production function can be expressed as follows:

$$q = \text{Min}\left(\frac{z_1}{a}, \frac{z_2}{b}\right)$$

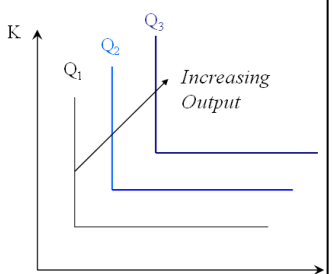
Where, q = quantity of output produced

Z_1 = utilized quantity of input 1

Z_2 = utilized quantity of input 2

a and b = constants

- Q = min{aK, bL}
- Capital and labor are perfect complements and cannot be substituted (no MRTS <=> no slope)
- Capital and labor are used in fixed-proportions
- Both inputs needed to produce output



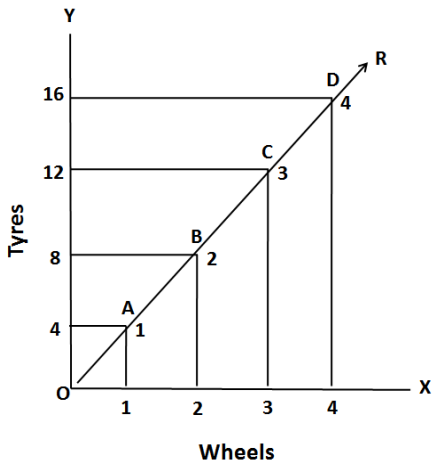
For example, tyres and steering wheels are used for producing cars. In such a case, the production function can be as follows:

Q = min (z_1/a, Z_2/b)

Q = min (number of tyres used, number of steering wheels used).

Suppose that the inputs "tires" and "steering wheels" are used in the production of a car (for simplicity of the example, to the exclusion of anything else). Then in the above formula *q* refers to the number of cars produced i.e., one in our example, z_1 refers to the number of tires used, and z_2 refers to the number of steering wheels used. Assuming that each car is produced with 4 tires and 1 steering wheel, the Leontief production function is

Number of cars = Min{¼ times the number of tires, 1 times the number of steering wheels}.



In the above given figure, OR shows the fixed Tyres-Wheels ratio, if a firm wants to produce 1 car, then 4 tyres and 1 wheel must be used.

Similarly, for the production of 3 cars and 4 cars, 12 tyres and 3 wheel and 16 tyres and 4 wheel must be employed respectively.

3. CES Production Function

Definition: The **Constant Elasticity of Substitution Production Function or CES** implies, that any change in the input factors, results in the constant change in the output. In CES, the elasticity of substitution is constant and may not necessarily be equal to one or unity.

In constant elasticity of substitution production function, all the input factors are taken into the consideration such as raw material, technology, labor, capital, etc. The marginal product of one factor increases with the increase in the value of the other factors of production. Also, the marginal product of labor and capital will be positive in case of constant returns to scale.

The constant elasticity of substitution production function can be expressed algebraically as:

$$\boldsymbol{Q = A[\alpha K^{-\beta} + (1-\alpha)L^{-\beta}]^{-1/\beta}}$$

Where, Q = output, K = Capital and L = Labor

A = efficiency parameter that shows the organizational aspects of production and the state of technology.

The Constant elasticity of substitution production function shows, that any change in the technology or organizational aspects, the production function changes with a shift in the efficiency parameter.

α= distribution parameter or capital intensity factor coefficient concerned with relative factors in the total output.

β = substitution parameter, that determines the elasticity of substitution

The homogeneity of the production function can be determined by the value of the substitution parameter (β), if it is equal to one, then it is said to be linearly homogeneous i.e. the proportionate change in the input factors results in the increase in the output in the same proportion.

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For example:

Consider the following production function.

$$\mathbf{Q = 5K + 10L}$$

It means 1 unit K produces 5 units and 2 units of L produce 10 units.

If we assume that, initially K = 1, L = 2, the total output may be obtained by substituting 1 for K and 2 for L in the below equation. Thus,

$$\mathbf{Q_{x1} = 5(1) + 10(2)}$$

$$\mathbf{25 = 5 + 20}$$

when inputs are doubled (i.e., K =2 and L = 4). Then,

$$\mathbf{Q_{x2} = 5(1x2) + 10(2x2)}$$

$$\mathbf{50 = 10 + 40}$$

Thus, the given production functions, when inputs are doubled, the output is also doubled.